There are more than 60,000 species of **vertebrates**, animals with a backbone. This is a relatively small number compared to, say, the 1 million insect species on Earth. But what vertebrates lack in number of species, they make up for in *disparity*, varying enormously in characteristics such as body mass. Vertebrates include the heaviest animals ever to walk on land, plant-eating dinosaurs that were as massive as 40,000 kg (heavier than 13 pickup trucks). The biggest animal ever to exist on Earth is also a vertebrate—the blue whale, which can exceed 100,000 kg. On the other end of the spectrum, the fish *Schindleria brevipinguis* is just 8.4 mm long and has a mass roughly 100 billion times smaller than that of a blue whale.

In this chapter, we will track the origin and evolution of the vertebrate body plan, from a notochord to a head to a mineralized skeleton to limbs with digits (such as toes). We'll also explore the major groups of vertebrates (both living and extinct), as well as the evolutionary history of our own species—*Homo sapiens*.

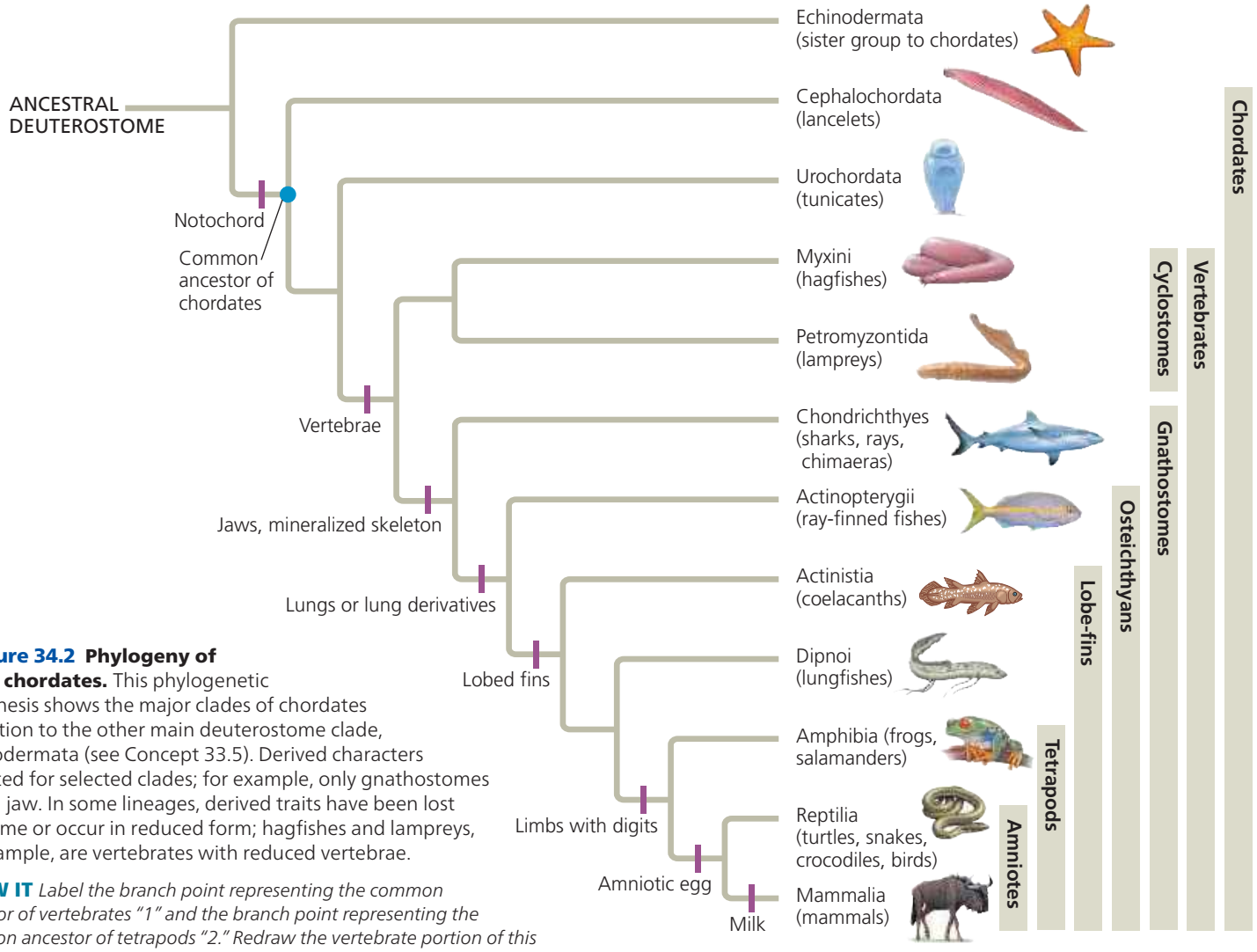
CONCEPT 34.1

Chordates have a notochord and a dorsal, hollow nerve cord

Vertebrates are members of the phylum Chordata, the chordates. **Chordates** are bilaterian (bilaterally symmetrical) animals, and within Bilateria, they belong to the clade of animals known as Deuterostomia (see Figure 32.11). As shown in **Figure 34.2**, there are two groups of invertebrate deuterostomes that are more closely related to vertebrates than they are to other invertebrates: the cephalochordates and the urochordates. Thus, along with the vertebrates, these two invertebrate groups are classified within the chordates.

Derived Characters of Chordates

All chordates share a set of derived characters, though many species possess some of these traits only during embryonic



▶ Figure 34.2 Phylogeny of living chordates. This phylogenetic hypothesis shows the major clades of chordates in relation to the other main deuterostome clade, Echinodermata (see Concept 33.5). Derived characters are listed for selected clades; for example, only gnathostomes have a jaw. In some lineages, derived traits have been lost over time or occur in reduced form; hagfishes and lampreys, for example, are vertebrates with reduced vertebrae.

DRAW IT *Label the branch point representing the common ancestor of vertebrates "1" and the branch point representing the common ancestor of tetrapods "2." Redraw the vertebrate portion of this tree by rotating its branches around these two branch points. Use the redrawn tree to explain why it is* not *correct to represent evolution as a sequence of events "leading to" humans and other mammals.*